Methods for Leakage Calculation
Supporting Document `IndigoCarbon_US-1_2023_0071`

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1 Overview

This document outlines the methods used to determine any leakage from the project. Potential sources of leakage include that from yield decline within the project scenario and from displacement in grazing to an area outside the project.

When accounting for grazing displacement, the average animal grazing days (AGDs) were compared between the project and baseline scenario. To account for yield decline, this document describes the calculation of actual production history (APH) for all fields. These calculated APH values are used to determine leakage emissions for the Project, in accordance with SEP v1.1, Section 5.5. This method was developed because APH is a difficult value for Field Managers to provide in all cases and there were well established methods to calculate this metric directly for all Field Managers. The core logic described below comes from an Extension document outlining the process for APH calculation[1]. That document can be found at:

https://www.extension.iastate.edu/agdm/crops/html/a1-55.html

2 Grazing Method Description

To prevent leakage from livestock displacement in accordance with CAR SEP section 5.5.1, the average AGD per animal type is calculated and compared between the project and baseline scenarios. The AGD per animal type for the project scenario was then set to the larger of the two values. This ensures that any livestock displacement does not lead to reduced emissions in the project scenario as required by SEP section 5.5.1.

3 APH Method Description

3.1 General Rules

- APH was calculated for each management zone and harvest of a cash crop

  - Each *crop growing season* (defined in IndigoCarbon_US-1_2023_0016) typically has one harvest of a cash crop.
  - Management zones are discussed in Section 6.1 of the Monitoring Plan.
- Every field was calculated in the same way for consistency.
- Yields entered by the Field Manager are used to calculate APH
  - When yield was not provided by a Field Manager or was outside of the QAQC ranges described in `IndigoCarbon_US-1_2023_0032`, yields from the National Agricultural Statistics Service (NASS; see `IndigoCarbon_US-1_2023_0069` for details) were used as described below.
- A "crop" in the leakage and APH calculation is a pair of: (1) a cash crop and (2) whether irrigation was done after planting and before the harvest (called "irrigation status" for short). Examples would include:
  - Irrigated corn
  - Non-irrigated soybeans
- All APH calculations were done at the management zone level before aggregating to the project level to calculate overall leakage, following equation 5.30 in SEP v1.1.

3.2 Specific Logic Implemented

APH in the carbon program in the project scenario

- APH is calculated for each harvest event (paired with an associated planting event).
- APH is an average over as few as four, and as many as ten, observations of yield for the same crop and irrigation status.
- To assemble those observations of yield for a particular harvest event, we look for yield records preceding the harvest event of interest that satisfy all the following:
  - (1) have the same crop and irrigation status
  - (2) were harvested in a calendar year less than or equal to 10 years prior to the calendar year of the harvest event of interest
- The yield in the harvest event of interest is included in the average. It is typically available, unless QA/QC flags the yield as out-of-range (in which case the yield is null and treated as if it were missing). More details on out-of-range yield are given in the section "Out of Range Yield Values" further down in this document.
- Yield observed in a baseline scenario can fall into the moving average for APH for a harvest event in the project scenario.
- NASS 10-year averages, called "NASS transitional yield" and denoted by $\text{Yield}_{\text{NASS}}$ below, are used to fill in missing data in both the project and baseline scenarios. The 10-year average uses the finest-scale NASS data available in every year, and the time window for the 10-year average ends on the calendar year of the harvest event of interest. If two or more observations of yield are missing, then the NASS data used to impute them must suffer penalties as described in the scenarios below.
  - If there are 4 or more yield values available from the grower, APH is a simple average, and no missing-data penalty is applied. For example, with 4 yield values (denoted by Yield_0, Yield_{-1}, Yield_{-2} Yield_{-3} Yield_{-4}), APH is:
    - APH = (Yield_0+ Yield_{-1} + Yield_{-2} + Yield_{-3}) / 4
  - If there are fewer than 4 yield values available, missing values (NULL or out of range) must be filled in with NASS data as described below. As noted above, $\text{Yield}_{\text{NASS}}$ is 10-year average yield according to NASS surveys, with the time window ending in the same year as the yield record Yield_0.
    - If there are 3 yield values, APH is an average of those three values and a NASS transitional yield:

APH= (Yield_0 + Yield_{-1} + Yield_{-2} + $\text{Yield}_{\text{NASS}}$) / 4

Table 1: Calculating APH yield would be calculated for a farm in a county with the transitional yield of 160 bushels/acre corn yields.

Calculating APH Yield					
	No Records	**One Year of Records**	**Two Years of Records**	**Three Years of Records**	**Four Years of Records**
Year One	160 bu. x 65% =104 bu	208 bu.	208 bu.	208 bu.	208 bu.
Year Two	160 bu. x 65% = 104 bu	160 bu. x 80% = 128 bu.	157 bu.	157 bu.	157 bu.
Year Three	160 bu. x 65% =104 bu.	160 bu. x 80% = 128 bu.	160 bu x 90% = 144 bu	138 bu.	138 bu.
Year Four	160 bu x 65% 104 bu.	160 bu. x 80% = 128 bu.	160 bu x 90% = 144 bu	160 bu. x 100% = 160bu.	177 bu.
APH Yield	104 bu	148 bu.	163 bu.	166 bu.	170 bu.

- If there are 2 yield values, APH is an average of those two values and 90% of 2 NASS T yields:

  $\text{APH} = (\text{Yield}_0 + \text{Yield}_{-1} + (\text{Yield}_{\text{NASS}} * 0.90) + (\text{Yield}_{\text{NASS}} * 0.90)) / 4$

- If there is just one yield value available, APH is an average of that yield and 80% of 3 NASS transitional yields.

  $\text{APH} = (\text{Yield}_0 + (\text{Yield}_{\text{NASS}} * 0.80) + (\text{Yield}_{\text{NASS}} * 0.80) + (\text{Yield}_{\text{NASS}} * 0.80)) / 4$

- If no yield values are available (NULL or out of range) and there is no baseline yield data, APH is an average of 65% of NASS transitional yields

  $\text{APH} = ((\text{Yield}_{\text{NASS}} * 0.65) + (\text{Yield}_{\text{NASS}} * 0.65) + (\text{Yield}_{\text{NASS}} * 0.65) + (\text{Yield}_{\text{NASS}} * 0.65)) / 4$

- An example from the ISU extension document is shown in table 1
- Similarly, APH is calculated for regional yield using the process above with one key difference: Regional APH values never have a penalty applied because NASS yield estimates are available (i.e., records are not missing).

Resolution of NASS yield used

- NASS data is reported at a hierarchy of levels that nest within one another:
  - County
  - Agricultural Statistical District (ASD), which is a collection of counties within a state
  - State
  - National
- Yield is retrieved from NASS at the finest scale available (at the county level, else at the ASD level, else at the state level else at the national level) at which we either find data for the given irrigation status (called "irrigation matched") or we find data that is representative of irrigation status in that region (called "irrigation unmatched"). Thus, the overall hierarchy in decreasing order of preference was:
  - County / irrigation matched

  1. County / irrigation unmatched
  2. ASD / irrigation matched
  3. ASD / irrigation unmatched
  4. State / irrigation matched
  5. State / irrigation unmatched
  6. National / irrigation matched

7. National / irrigation unmatched

- If NASS showed multiple estimates for the same crop at the same unit, the mean value was used.

Out of Range Yield Values

- Yield values outside of our QAQC ranges were not gap-filled with extrapolation or default values as they weren't a model input. (See IndigoCarbon_US-1_2023_0031 for details on gap-filling with extrapolation or default values.) They were treated as NULL for all leakage purposes.
  - Since these values were treated as missing, fields with out-of-range values received heavier penalties to APH calculations, making this approach conservative.
  - See IndigoCarbon_US-1_2023_0032 for more details on QC ranges applied to yields.

Aggregating APH values across the project

- To compute the average APH values in Equations 5.31 and 5.32 in SEP v1.1 ($\overline{APH_{c,t}}$, $\overline{APH_{RA,c,t}}$, $\overline{APH_{c,hy}}$, and $\overline{APH_{RA,c,hy}}$), we compute an area-weighted average of all APH values for a particular crop, irrigation status, and scenario (project or baseline). The averages for the project scenario, $\overline{APH_{c,t}}$ and $\overline{APH_{RA,c,t}}$, are computed from all APH values for cultivation cycles assigned to the current monitoring report.

4 Works Cited

1. Iowa State University Extension and Outreach, *Proven Yields and Insurance Units for Crop Insurance*, 2017, https://www.extension.iastate.edu/agdm/crops/html/a1-55.html

5 Summary of Changes

Section	Date	Summary of Changes
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